

DAX MEASURES DOCUMENTATION

Project Title: Telco Customer Churn Analysis

Programme: AnalystLab Africa Data Analytics Internship

Week 3 Project

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Introduction

This document provides an overview of the DAX (Data Analysis Expressions) measures created for the Telco Customer Churn Analysis Dashboard.

The measures were developed in Power BI.

DAX Measures.

1. Total Customers

Dax formula: Total Customers = Distinctcount ('Telco Customer Churn'[CustomerID])

Purpose: Counts unique customers in the dataset

Business Use: Monitors overall size of the customer base.

2. Churned Customers

Dax formula: Churned Customers = Calculate ([Total Customers],'Telco Customer Churn'[Churn] = "Yes")

Purpose: Counts customers who have discontinued services.

Business Use: Quantifies customer losses and retention challenges.

3. Churn Rate

Dax Formula: Churn Rate = DIVIDE ([Churned Customers],[Total Customers],0)

Purpose: Calculates percentage of customers who churned.

Business Use: Monitors overall customer attrition level.

4. Retention Rate

Dax formula: Retention Rate = 1 - [Churn Rate]

Purpose: Calculates percentage of customers retained.

Business Use: Evaluates customer retention performance.

5. Active Customers

Dax formula: Active Customers = [Total Customers] - [Churned Customers]

Purpose: Counts customers who have not churned.

Business Use: Measures the current retained customer base.

6. Average Tenure

Dax formula: Average Tenure = AVERAGE ('Telco Customer Churn'[Tenure])

Purpose: Calculates average customer relationship duration.

Business Use: Assesses customer longevity and loyalty.

7. Average Monthly Charges

Dax formula: Average Monthly Charges = AVERAGE ('Telco Customer Churn'[MonthlyCharges])

Purpose: Calculates average monthly customer charges.

Business Use: Measures average monthly customer value.

8. Total Revenue

Dax formula: Total Revenue = SUM ('Telco Customer Churn'[TotalCharges])

Purpose: Calculates total recorded customer charges.

Business Use: Assesses overall financial value of customers.

9. Revenue Lost

Dax formula: Revenue Lost = CALCULATE ([Total Revenue], 'Telco Customer Churn'[Churn] = "Yes")

Purpose: Calculates charges associated with churned customers.

Business Use: Estimates financial exposure from customer churn.

10. Estimated Customer Lifetime Value

Dax formula: Estimated Customer Lifetime Value = [Average Monthly Charges] * [Average Tenure]

Purpose: Estimates average customer value over tenure.

Business Use: Supports customer retention value assessment.

11. Revenue Impact from Churn

Dax formula: Revenue Impact from Churn = -[Revenue Lost]

Purpose: Represents churn related revenue as negative impact.

Business Use: Highlights financial consequences of customer churn.